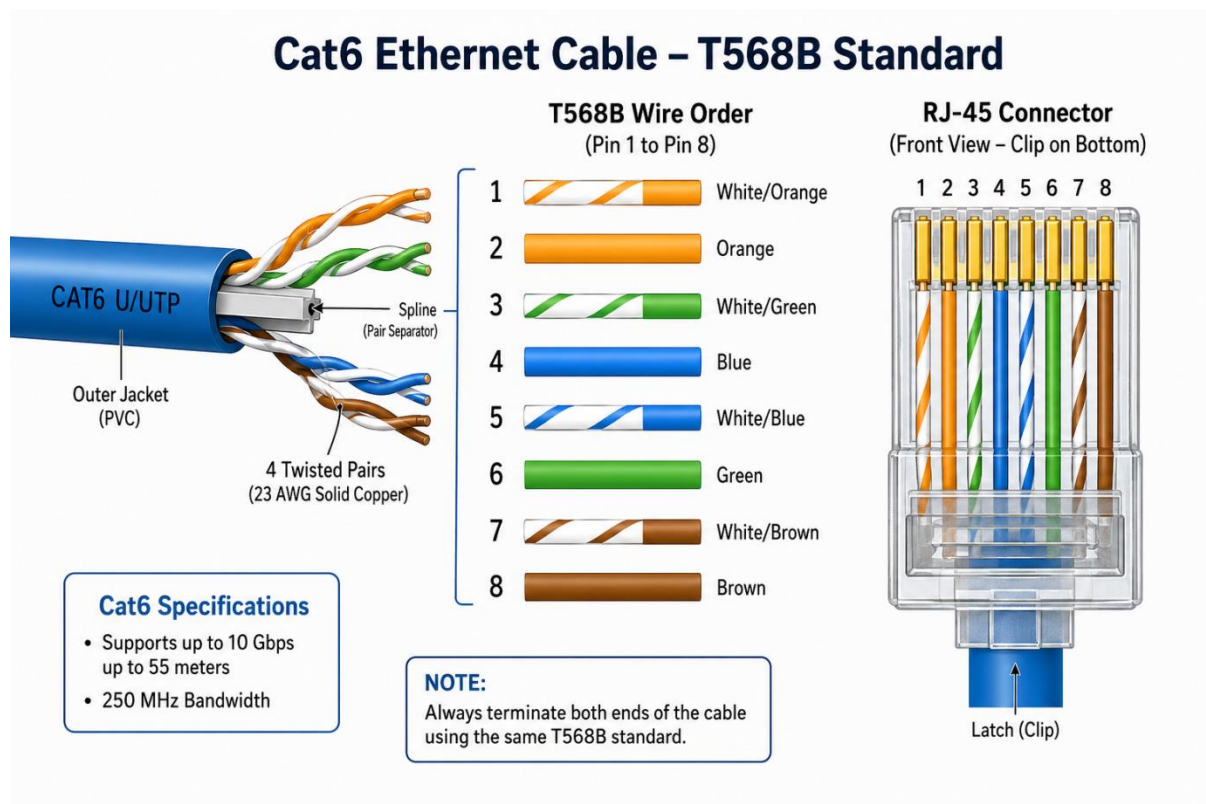


1. Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labelling the RJ-45 connector pins.



2. List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a Wi-Fi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

a) Cat 5e – Wi-Fi router

- A Cat5e cable is commonly used to connect a Wi-Fi router to a modem or switch. It supports up to 1 Gbps, making it suitable for home and home and small-office internet connections at a reasonable cost.

b) Cat 6 – Zomato POS terminal

- A Cat6 cable provides faster speeds and better protection against interference than Cat5e. This helps a Zomato POS terminal maintain a reliable, fast connection for processing orders and payments in busy restaurants.

c) Fiber Optic Cable – IPL streaming server

- Fiber optic cables transmit data using light, allowing very high bandwidth and long-distance communication with minimal signal loss. This makes them ideal for IPL streaming servers, which must deliver high-quality live video to millions of viewers.

3. Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

- I watched a YouTube tutorial demonstrating how to crimp an RJ-45 connector onto an Ethernet cable and summarized the process in my words.
- **Step-by-step process:**
- Gather the required tools: an Ethernet cable, RJ-45 connector, a cable stripper a crimping tool and ideally a cable tester.
- Strip about 2.5-3cm of the outer cable jacket without damaging the inner wires.
- Untwist the four wire pairs and straighten the eight individual wires.
- Arrange the wires in the correct order which is White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown and Brown.
- Flatten the wires and trim them so that all eight ends are even and the correct length.
- Insert the wires fully into the RJ-45 connector, making sure each wire reaches the end of its channel and that the outer cable jacket extends slightly inside the connector for strain relief.
- Place the connector into the crimping tool and squeeze the handles firmly until the crimp is complete.
- Test the finished cable with a cable tester to verify that all eight conductors are connected correctly and in the proper order.
- **Two common mistakes to avoid:**
- Using the wrong wire order or mixing up the conductors. Even a single wire out of sequence can prevent the cable from working properly.
- Not pushing the wires fully into the connector before crimping. If the wires do not reach the front contacts or the cable jacket is not secured inside the connector, the connection may be unreliable or fail completely.

4. Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

Device Connection	Recommended Cable	Connector Cable	Why?
Gaming console, TV/Projector	HDMI 2.1 cable	HDMI Type-A	Supports upto 4k/120Hz or 8k, HDR and surround sound with minimal latency. Ideal for modern consoles like Playstation5 and Xbox Series X.
PC, Gaming monitor	DisplayPort 1.4/2.1 cable	DisplayPort	Provide high refresh rates and excellent gaming performance.
Network for consoles/PCs	Cat6 Or Cat6a Ethernet cable	RJ-45 connector	Offers reliable, high-speed wired internet, reducing lag during online gaming.
Audio system, AV receiver	HDMI Earc or Optical	HDMI/Optical connector	Delivers high-quality digital audio to speakers or soundbars.
Controllers or accessories	USB Type-C	USB-C connector	Fast charging and data transfer for modern gaming.

5. Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a Wi-Fi router or at a café's POS counter), and describe what type it is and what devices it connects.



- Here, Ethernet cable is connected to a Wi-Fi router.
- When it connects, one end plugs into the LAN or WAN port of Wi-Fi router.
- The other end typically connects to a switch, desktop computer or another networking device.
- It carries wired network data, providing a fast and reliable internet or local network connection.